Experiment 06

Logistic Regression – Binary Classification

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# Experiment 06: Logistic Regression Binary Classification

# 1. Import libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import confusion_matrix
from sklearn.metrics import accuracy_score, precision_score
from sklearn.metrics import recall_score, f1_score
from sklearn.metrics import classification_report

# 2. Load Breast Cancer dataset
data = load_breast_cancer()

X = data.data
y = data.target

# 3. Convert dataset into DataFrame
df = pd.DataFrame(X, columns=data.feature_names)
df['target'] = y

# 4. Display first five records
print("First Five Records:")
print(df.head())

# 5. Check dataset information
print("\nDataset Information:")
print(df.info())

# 6. Check missing values
print("\nMissing Values:")
print(df.isnull().sum())

# 7. Split data into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)

# 8. Feature Scaling
scaler = StandardScaler()

X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

# 9. Create Logistic Regression model
model = LogisticRegression(max_iter=1000)

# 10. Train the model
model.fit(X_train, y_train)

# 11. Predict the output
y_pred = model.predict(X_test)

print("\nPredicted Values:")
print(y_pred)

# 12. Calculate Confusion Matrix
cm = confusion_matrix(y_test, y_pred)

print("\nConfusion Matrix:")
print(cm)

# 13. Calculate Accuracy
accuracy = accuracy_score(y_test, y_pred)
print("\nAccuracy:", accuracy)
# 14. Calculate Precision
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precision = precision_score(y_test, y_pred)
print("Precision:", precision)

# 15. Calculate Recall
recall = recall_score(y_test, y_pred)
print("Recall:", recall)

# 16. Calculate F1 Score
f1 = f1_score(y_test, y_pred)
print("F1 Score:", f1)

# 17. Classification Report
print("\nClassification Report:")
print(classification_report(y_test, y_pred))

# 18. Visualize Confusion Matrix
plt.figure(figsize=(6, 4))

sns.heatmap(
    cm,
    annot=True,
    fmt='d',
    cmap='Blues',
    xticklabels=['Malignant', 'Benign'],
    yticklabels=['Malignant', 'Benign']
)

plt.xlabel("Predicted Values")
plt.ylabel("Actual Values")
plt.title("Confusion Matrix - Logistic Regression")
plt.show()
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